

Zero-phonon lines, electron-phonon coupling, and optically induced electron spin polarization of divacancies in *p*-type 4H-SiC: A low-temperature electron paramagnetic resonance study

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The zero-phonon lines, phonon sidebands and the dynamics of optically induced spin polarization of the neutral divacancies ($V_{Si}V_C$)⁰ in 4H-SiC have been investigated by EPR spectroscopy. Via *in-situ* resonant optical excitation at $T = 4$ K the zero phonon lines (ZPL) between the 3A_2 groundstate and the 3E excited states and their absorption phonon side bands (PSB) have been accurately determined for each of the four VV^0 centers (hh, kk, kh, hk) from the spectral dependence of the EPR line intensity. Their values, respectively 1130.7 nm (hh), 1129 nm (kk), 1107.5 nm (kh) and 1077.3 nm (hk) are in good agreement with previous PL studies and with theoretical predictions. Decomposition of the phonon sidebands of the basal VV centers reveals a coupling with phonons of 24, 36, and 47.5 meV (V_{kh}) and 11.5, and 18.6 meV (V_{hk}). The low-temperature optical excitation of the VV^0 centers gives rise to a 100% ground-state spin polarization which persists at $T = 4$ K for hundreds of seconds, when the excitation is switched off. Contrary to the case of VV centers in *n*-type nitrogen doped 4H-SiC, the VV centers in *p*-type doped 4H-SiC do not show any charge-state instability under optical excitation.

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I. INTRODUCTION

4H-SiC has been shown to be not only an important microelectronic material for high power devices but also an excellent quantum material for application in photonic networks and nanoscale sensing of magnetic fields and temperatures [1–6]. Whereas many of such applications have been realized previously with NV centers in diamond, it has only more recently been realized that SiC is a platform with superior material, optical and defect properties. They rely on a family of solid-state qubits, which can be intrinsic defects such as Simonov vacancies V_{Si} , divacancies $V_{Si}V_C$ or mixed $V_{Si}N_C$ (NV) centers for example. In particular VV^0 [7] and NV^- [8,9] centers have been extensively studied. Transition metal related ($S = 1$) defects such as Vanadium V^- [10] are also promising, but have been less investigated.

These defects, suitable for qubit application, have been studied in bulk samples, thin epitaxial layers, and photonic structures [1–6]. The versatility of these defects is related to the mature control of the electrical properties of SiC, which can be easily prepared as *n*-type conductive (Nitrogen doped), *p*-type conductive (Aluminum doped), or high resistive material. Initially these studies were performed on ensembles of defects, but the excellent optical properties of these centers, emitting in the NIR spectral range 0.8 μm , . . . , 1.3 μm , have also allowed single defect spectroscopy. More advanced technologies based on photonic cavity formation, detection via

waveguides and the use of mm large arrays of qubits have equally been reported; see, e.g., Ref. [1] for a review.

In addition to the intrinsic divacancies, $V_{Si}V_C$, two further divacancy related defects labeled PL5, PL6 have been reported and studied for their interesting qubit properties [11,12]. They have been initially attributed to stacking-fault related divacancies [13]. However very recently this assignment has been questioned and a new model, $V_{Si}O_C$, has been proposed [14]. Surprisingly, in the standard approach for formation of vacancy defects, i.e., particle irradiation of initially *n*-type or *p*-type material followed by high-temperature annealing at 900 °C, the PL5/PL6 centers are not observed by EPR spectroscopy. This result is in good agreement with a recent new model assigning them to an oxygen-vacancy complex. A drawback, their association with oxygen does not allow to introduce them in a controlled way.

In this study, we have investigated the optical and spin properties of neutral divacancy defects $V_{Si}V_C$ in 4H-SiC by resonantly excited photo-EPR spectroscopy. The crystal structure of 4H-SiC gives rise to four different first nearest neighbor divacancy centers due to the presence of two distinct lattice sites, labeled quasicubic (k) and hexagonal (h), respectively. The resulting divacancy configurations are $V_{Si}(h)V_C(h)$, $V_{Si}(k)V_C(k)$, $V_{Si}(k)V_C(h)$, and $V_{Si}(h)V_C(k)$ labeled VV_{hh} , VV_{kk} , VV_{kh} , and VV_{hk} . The point-group symmetries of the axial (hh, kk) and basal (kh, hk) divacancies are C_{3v} and C_{1h} , respectively. Their ground states, 3A_2 for the axial centers and $^3A'$ (hk), $^3A''$ (kh) for the basal centers [28], are characterized by their different spin properties (*g* tensor), zero-field splitting, hyperfine interactions and defect multiplicities. Similar to the NV^- center in diamond, but in the spectral range

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TABLE I. VV° in $4H$ -SiC, experimental results of the g tensor, zero-field splitting parameters D , E , the ZPL lines determined experimentally and theoretically in this work as well as the previously reported ZPL of the four VV centers (PL1 – PL4) determined from PL spectroscopy; the last line shows the (+/0) charge transition levels of the VV° centers with respect to the valence band maximum (VBM). Note that the (0/-) CTL, relevant in n -type material are found around 2.0 eV above the VBM, and thus 1.2 eV below the CBM [29]. The major principal axes $g_{//}$ of the experimental g tensor of all VV centers are along the respective defect axes.

	Spin Properties				This work
	VV _{hh}	VV _{kk}	VV _{kh}	VV _{hk}	
<i>g</i> tensor <i>g</i> _⊥ , <i>g</i> _∥	2.0020 2.0038	2.0020 2.0038	2.0020 2.0038	2.0020 2.0038	Expt. theory
<i>g</i> _⊥ (<i>g</i> ₁), <i>g</i> _∥ <i>g</i> ₂	2.0024 2.0038	2.0026 2.0036	(2.0027) 2.0033 (2.0027)	(2.0027) 2.0034 (2.0028)	
D (MHz)	1336	1305	1208	1313	Expt. theory
	1322	1305	1135	1246	
E (MHz)	0	0	82	37	Expt. theory
			89	46	
Optical Properties					
	PL1	PL2	PL3	PL4	
ZPL1 (nm)	1130.7	1129.5	1107.5	1077.3	Expt. This work
ZPL2 (nm)	1127.0	1126.0	1102.0	1073.5	This work
ZPL1 (eV)	1.0965	1.0977	1.1195	1.1509	This work
ZPL2 (eV)	1.1001	1.1011	1.1251	1.1550	This work
ZPL (eV)	1.095	1.096	1.119	1.150	Falk [7]
ZPL (eV)	1.079	1.081	1.062	1.100	Theory r ² SCAN [28]
	1.056	1.044	1.081	1.103	HSE [20]
	1.051	1.039	1.060	1.091	PBE, vert.* (*This work)
Charge Transition Levels (eV)					
VV (+/0)	1.070	1.010	1.051	1.081	[29]

of 1.1 μm , their optical properties are dominated by radiative recombination between the 3A_2 -like ground state and the first 3E -like excited triplet state. As has been shown before [15,16], resonantly excited EPR spectroscopy allows a selective, center specific study of the related zero-phonon line (ZPL) absorption, $^3A_2 \rightarrow ^3E$ (for the axial centers hh,kk), $^3A' \rightarrow ^3A'$ and $^3A'' \rightarrow ^3A''$ (for the basal pairs hk and kh, respectively), with very high spectral resolution. One further decisive advantage of this approach is that the selective excitation avoids the complications resulting from the overlap of the different absorption bands in photo-luminescence excitation spectroscopy.

The spin properties of the VV centers and the photoluminescence spectra of the PL centers in $4H$ -SiC have been studied previously experimentally and by first principle calculations [7,17–21]. Contrary to most of the previous studies performed on n -type conducting $4H$ -SiC we have used in this study p -type, Al doped $4H$ -SiC as the starting material. As we will show below this has important advantages for the charge-state stability of the centers. We investigated the absorption spectra of each VV center close to the ZPL, which in addition to the direct association with the PL centers allows the analysis of the excited state configuration as well as the phonon coupling. We have further investigated the dynamics of the ground-state spin polarization at $T = 4$ K and their decline after the photoexcitation.

II. COMPUTATIONAL METHODS

First-principles calculations were performed in the framework of density functional theory (DFT) using the Quantum ESPRESSO package [22,23]. For modeling the defect structures, we used a hexagonal 432-atom supercell (with experimental lattice constants $a = 3.073$ Å, $c = 10.053$ Å), periodic

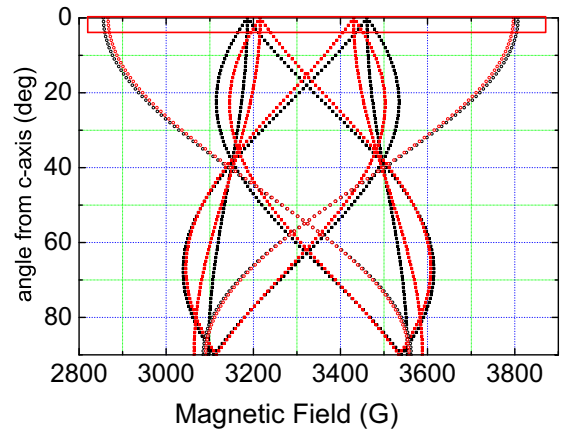


FIG. 1. Angular variation of the resonance fields of the four VV centers for a rotation of the magnetic field in the [1100] plane; the spectrum for $B//c$ is indicated.

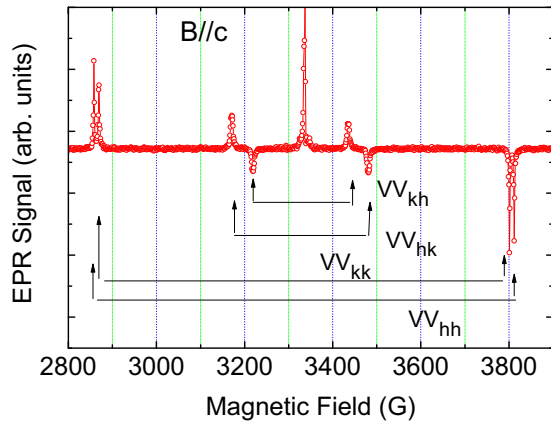


FIG. 2. EPR spectrum at $T = 4$ K and $B//c$ under optical excitation with 808 nm, which shows the four VV centers with a 100% spin polarization.

boundary conditions and a shifted $2 \times 2 \times 2$ k -point sampling. Effects of exchange and correlation are taken into account by using the semilocal PBE functional, norm-conserving pseudopotentials, and a plane-wave basis set with 680 eV energy cutoff. For the calculation of the zero-phonon lines (ZPL) we apply the same procedure (via vertical excitation energies) used successfully for the NV centers in 4H-SiC [15] and 6H-SiC [24]. Since the approach is based on the PBE functional (with underestimated fundamental gap of 2.43 eV), it profits systematically from error cancellations and thus gives reliable estimates for the relative values for the ZPL energies of the different VV configurations [15,24]. Based on fully relaxed ground-state geometries (forces below 0.01 eV/Å), we calculated the magnetic signature of the spin-triplet state of the divacancies. To obtain accurate estimates for the g tensors and both, the D as well as E value of the zero-field splitting (ZFS), we used scalar relativistic, norm-conserving gauge-including (GI) PAW pseudopotentials and double-perturbation theory. In the calculations of the ZFS parameters, besides spin-spin contribution [21], the effect of

the spin-orbit interaction has been considered [25]. The dominating spin-spin part is due to dipole-dipole interaction of the two-particle density [26] and tends to be overestimated by the so-called spin contamination, a spurious kind of antiferromagnetic spin polarization, which in quantum chemistry is well known to result in overestimated absolute magnetic moments. In the context of ZFS-splitting, this effect is manifested by nonvanishing D -values in the nominally diamagnetic $m_s = 0$ sublevel. In order to correct for this artificial effect, for a better comparison with the experimental data, we evaluate the spin-spin contributions via $D_{ss} = D_{ss}(m_s = \pm 1) - D_{ss}^{BS}(m_s = 0)$ [21]. Here, the label BS reflects the broken symmetry (BS), introduced by assuming an antiferromagnetic coupling in the $m_s = 0$ state, which allows to quantify the required correction.

III. EXPERIMENTS

The p -type Al doped 4H-SiC bulk samples were irradiated with 2MV electrons and subsequently annealed at $T = 900$ °C for 30 min. After irradiation and annealing, the samples are electrically compensated and we observe no longer the neutral Al⁰ acceptor EPR spectrum. The typical sample size of these c -plane samples is 4×4 mm².

The EPR measurements were performed with a commercial X-band spectrometer with 100 kHz field modulation and lock-in detection. All measurements were done at low temperature $T = 4$ K. Due to the long spin-lattice relaxation times of all four VV centers, we were able to measure them at the lowest available microwave power of $P = 200$ nW. For the resonant optical excitation, we used a 200 W halogen lamp as light source and a Czerny-Turner monochromator with a resolution of 3 nm for spectral selection. The output of the monochromator was coupled to an optical fiber, which allowed *in situ* excitation of the samples.

The ground-state spin polarization of the VV⁰ centers is deduced from the amplitudes of the $m_s = 0/+1$ and $m_s = 0/-1$ transitions and their respective phases (in absorption and emission, respectively). The dynamics of the spin polarization

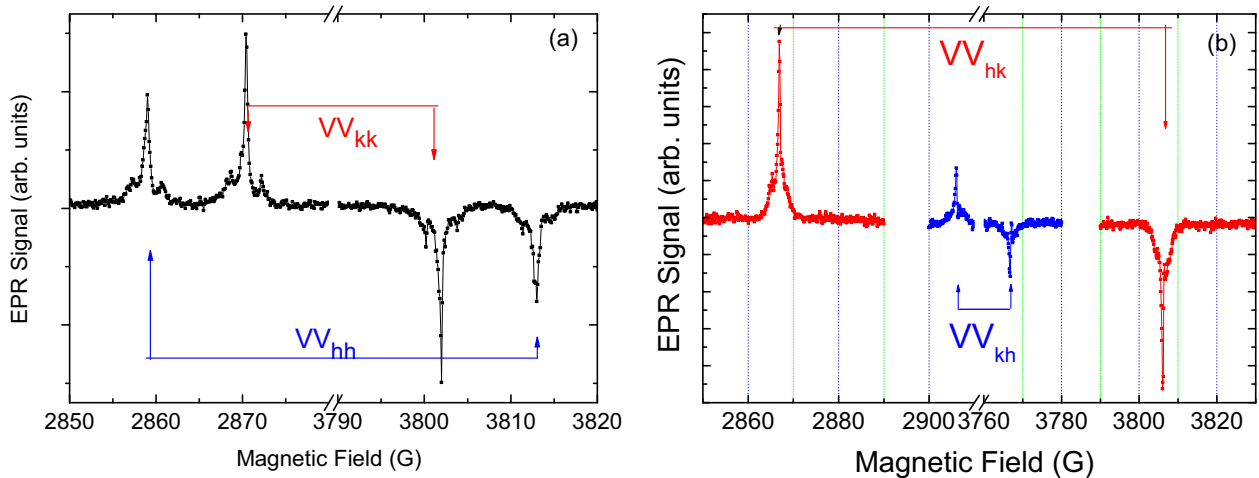


FIG. 3. High-resolution EPR spectrum of the axial divacancies VV_{hh}(a) and VV_{kk}(a) for $B//c$ and the basal divacancies VV_{kh}, VV_{hk}(b) for $B//70^\circ$ (w.r.t.c) at $T = 4.2$ K; they show the high-intensity central lines and the weak sidelines resulting from super hyperfine (SHF) interaction with neighboring ²⁹Si nuclei (with nuclear spin $I = 1/2$).

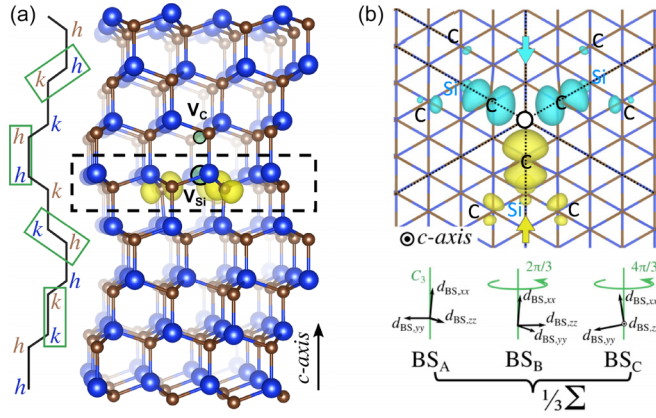


FIG. 4. (a) Crystal structure of 4H-SiC illustrating the two inequivalent Si (blue) and C (brown) lattice sites. Spin density (giving rise to the *direct* spin-spin D_{ss} values) exemplarily shown for the neutral VV center with axial hh configuration (yellow isosurface), predominantly localized in a region (dashed box) close to the V_{Si} vacancy. The other VV configurations are also indicated (left). (b) Better agreement with experiment requires “*spin-decontamination*” of the $m_S = 0$ sublevel via the three possible broken-symmetry (BS) configurations (bottom right) with antiferromagnetically coupled spin density (yellow spin-up, turquoise spin-down), exemplarily shown for BS_A (top right). The other two possibilities, BS_B and BS_C result by rotations with 120° and 240° with respect to the defect axis. The “*true low-spin*” $D(m_S = 0)$ tensor is defined as their average restoring the axial symmetry of the spin system (see text for further details).

effects were measured with excitation by a nonresonant laser (808 nm/100 mW) with a time resolution of 100 ms.

A. Results and discussion: EPR spectra

The EPR spectra of the VV centers are characterized by their Spin Hamiltonian $\mathbf{H}$ given by

$$\mathbf{H} = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S} + \mathbf{S} \cdot \mathbf{D} \cdot \mathbf{S} + \sum (\gamma_j \mathbf{B} \cdot \mathbf{I}_j + \mathbf{S} \cdot \mathbf{A}_j \cdot \mathbf{I}_j)$$

with electron spin $S = 1$, $\mathbf{g}$ the electronic g tensor, $\mathbf{D}$ the zero-field splitting tensor, and $\mathbf{A}$ the hyperfine interaction tensor.

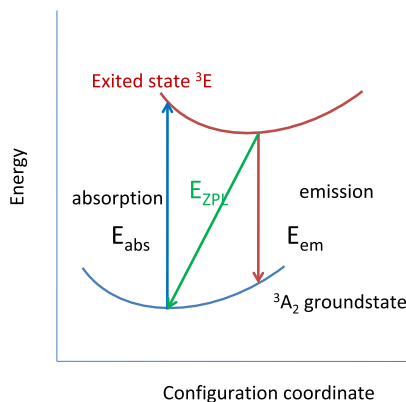


FIG. 5. Configuration coordinate diagram for axial pairs showing the ZPL and phonon-assisted transitions between the ground state 3A_2 and the doublet-split branches of the 3E excited state.

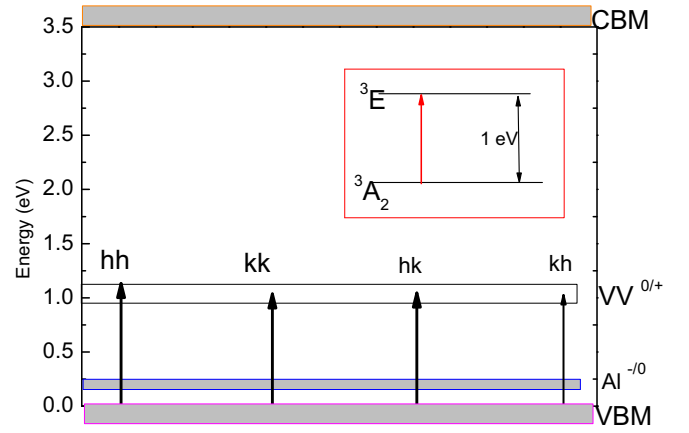


FIG. 6. Scheme of resonant excitation of the VV centers: $VV^+ \rightarrow VV^0 + h$; charge transition levels taken from Ref. [29]. The threshold for charge conversion is slightly above the energy of the (axial) $^3A_2 \rightarrow ^3E$ intracenter excitation of the VV^0 centers [29].

The g values and the zero-field splitting parameters D , E are given in Table I. Due to the distinct values and orientation of the $\mathbf{D}$ tensor, the VV centers can be easily studied individually.

In Fig. 1 we show the angular variation of the resonance fields of the four VV centers. For B//c the magnetic field is aligned parallel to the symmetry axis the two axial VV centers, whereas for the basal ones the magnetic field (B//c) is at 70° from their symmetry axis (Fig. 1). In Fig. 2 we show the corresponding EPR spectrum obtained at $T = 4$ K under nonresonant excitation with 808 nm laser. We observe simultaneously the four VV^0 centers, labeled VV_{hh} , VV_{kk} , VV_{kh} , and VV_{hk} . In addition, in the central part of the spectrum, we observe the EPR spectra of the positively charged carbon vacancy. One might have expected to observe equally the EPR spectra of the $1+$ charged VV centers. However this is not the case. Possible explanations are (i) the existence of a second charge transition level ($2+/+$), which would result for VV^{2+} in a EPR-silent, diamagnetic ($S = 0$) ground state, or (ii) an insufficient defect concentration. In this context it should be recalled that the high ground-state spin polarization enhances strongly the EPR signal intensity of VV^0 as compared to its

TABLE II. DFT-calculated ZFS parameters (D and E values in MHz) – *direct* spin-spin, corrected (“*spin-decontaminated*”) spin-spin, spin-orbit (D_{so}), and total (D_{ss} corrected + D_{so}) values. For basal pairs, the $E = \frac{1}{2}(D_{xx} - D_{yy})$ values are in parentheses. The attribution of the experimental spectra to the PL lines is clear thanks to the individual “*resonant excitation*” of the EPR transitions (Expt* from Ref. [7]).

	D_{ss} Direct	D_{ss} Corrected	D_{so}	D_{tot}	Expt.	Expt*
hh (PL1)	1698.5	1277.6	44.7	1322.3	1336	1336
kk (PL2)	1647.9	1234.4	70.4	1304.8	1305	1305
kh (PL3)	1592.3	1126.8	9.9	1134.7	1208	1222
	(107.9)	(89.3)	(0.3)	(89.6)	(82)	(82)
hk (PL4)	1656.0	1219.2	28.5	1246.0	1313	1334
	(42.2)	(45.0)	(0.2)	(45.8)	(37)	(19)

TABLE III. Time constants of the decline of the VV° by hole capture $VV^\circ + h \rightarrow VV^+$ at $T = 4$ K and the decline of the spin polarization P .

	VV_{hh} (PL1)	VV_{kk} (PL2)	VV_{kh} (PL3)	VV_{hk} (PL4)
Optically induced spin polarization (%)	100	100	100	100
Time constants(s) of the EPR signal decay (double exponential)	$t_1 = 6.4$ s $t_2 = 69$ s	$t_1 = 7$ s $t_2 = 118$ s	$t_1 = 4.1$ s $t_2 = 134$ s	$t_1 = 3.2$ s $t_2 = 114$ s
Time constant of the polarization decay (single exponential)	$t_c = 64$ s	$t_c = 109$ s		

thermal equilibrium value; this is not the case for VV^+ with a “normal” thermal polarization at $T = 4$ K.

As shown in Figs. 2 and 3, under 808 nm excitation, all four VV° centers are not only simultaneously excited but also 100% spin polarized. The degree of the spin polarization is deduced from the amplitudes and phases of the low-field and high-field magnetic dipolar transitions $m_s = 0/-1$ and $m_s = 0/+1$. It is defined as $P = (I_{hf} - I_{lf}) / (I_{hf} + I_{lf})$. For all four centers we observe that the low field and high field $m_s = 0/-1$, $m_s = 0/+1$ transitions are opposite in phase (preferential population of the $m_s = 0$ state) and of equal intensity, which corresponds to a spin polarization of 100%. The PL5 and PL6 centers, often named divacancy centers, are not observed in line with the recent attribution of PL5/PL6 to oxygen vacancy complexes [14].

The EPR parameters, g tensors and zero-field splitting (ZFS) parameters D , E determined at $T = 4$ K are shown in Table I for the four VV centers (hh, kk, kh, hk). They are in

good agreement with the previously published values [7,17] and also in reasonable agreement with our theoretical DFT modeling for the g tensor as well as for the D and E parameters of ZFS.

The deviation of the g values from the free-electron value $g_e = 2.002319$ is determined by spin-orbit coupling (SOC) [27]. Hence, the DFT-calculated g tensors reflect the different symmetry of the spin systems, C_{3v} for the axial configurations (hh, kk) and C_{1h} for the basal divacancies (hk, kh). In contrast to the axial centers (with degenerate $g_1 = g_2 = g_\perp$), the basal configurations thus provide three independent principal g values. For fitting the experimental EPR spectra, however, a common axial g tensor for all four divacancies is used in literature as well as in this work: $g_{//} = 2.0038$ and $g_\perp = 2.0020$ with respect to the corresponding defect axis. For the axial centers, in particular for the hh configuration, a very well agreement with the DFT-calculated values is found. For the basal divacancy configurations hk

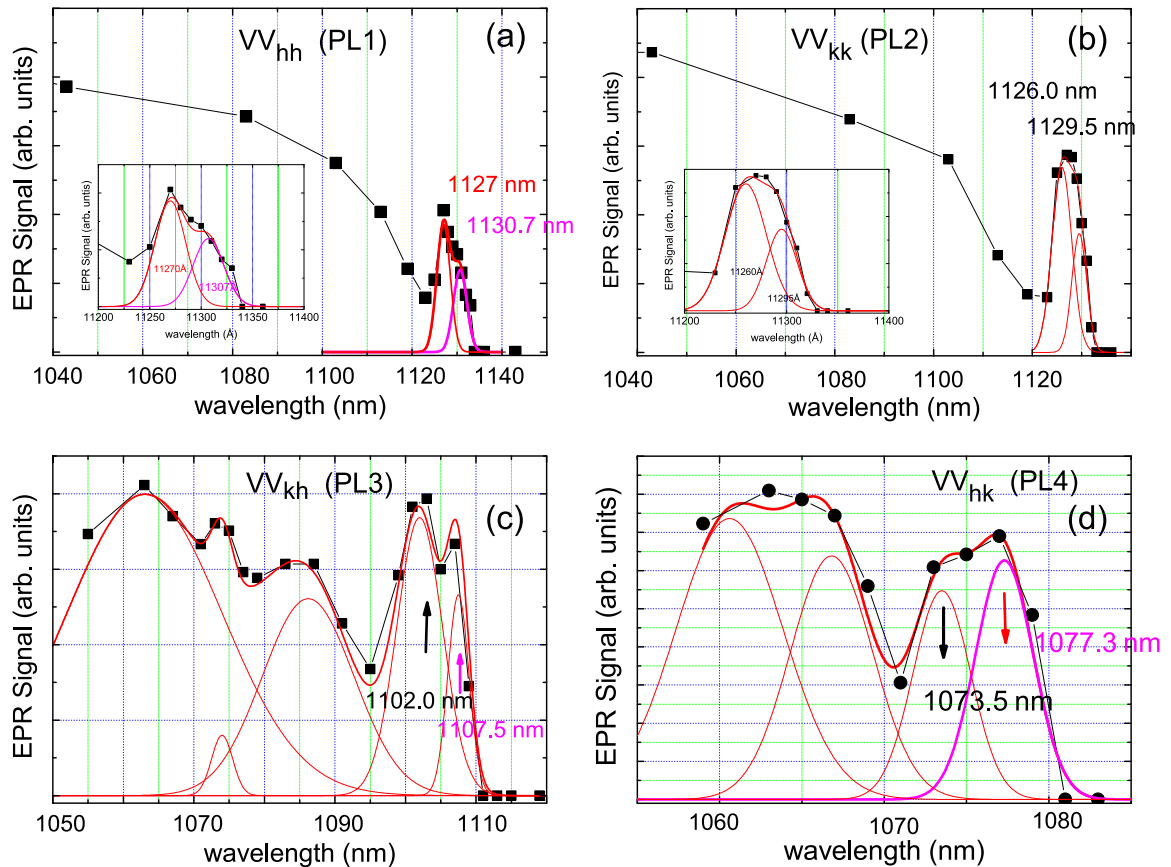


FIG. 7. Resonant excitation spectra ($^3A_2 \rightarrow ^3E$) of the four VV° centers at $T = 4$ K and their simulation with Gaussian line shapes [(a), (b), (c), (d)]. Both types of divacancies, axial (top) [(a) and (b)] as well as basal [(c) and (d)] (bottom) show a doublet splitting (see also Fig. 8).

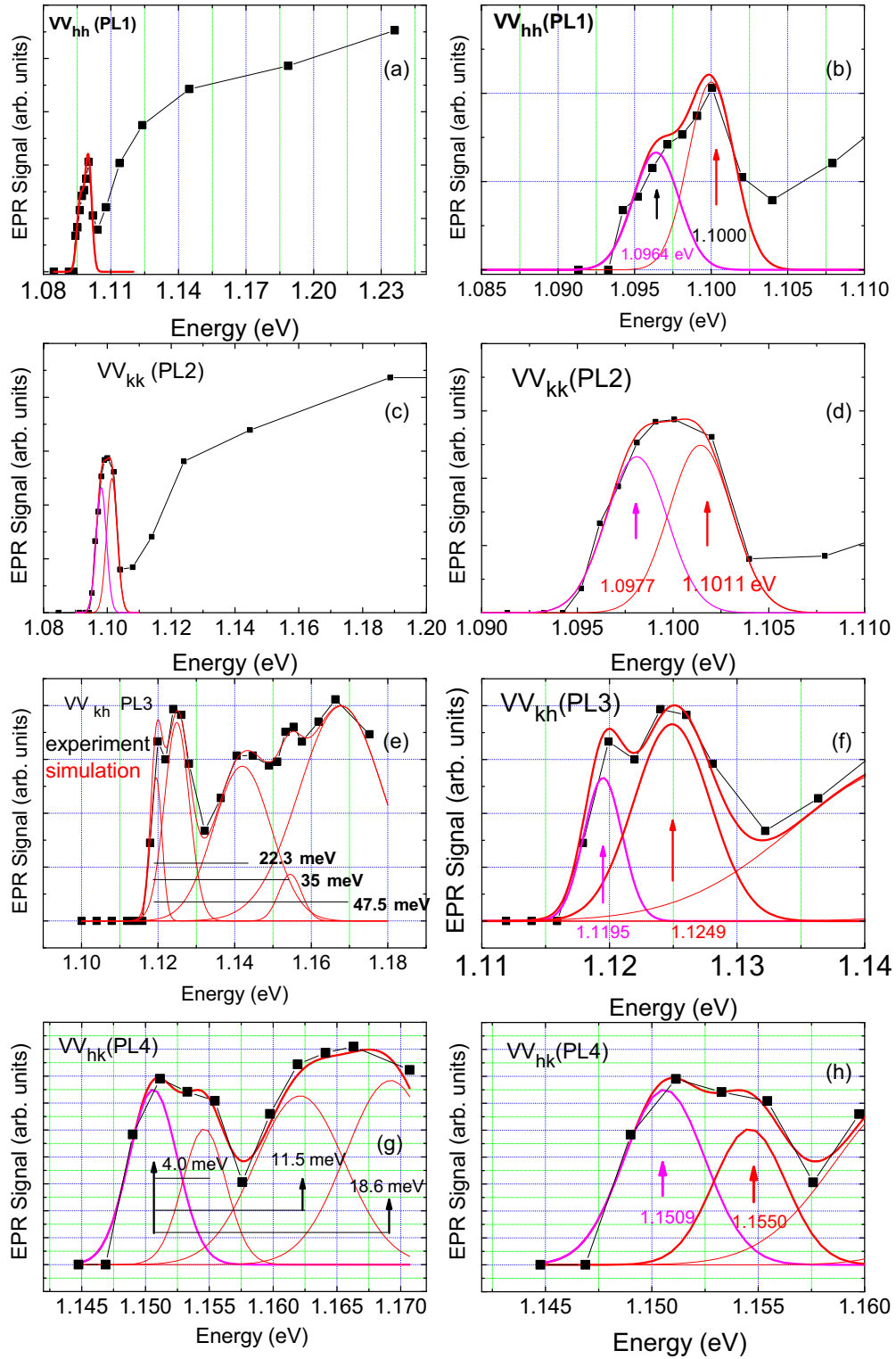


FIG. 8. Large scale resonant excitation spectra (${}^3A_2 \rightarrow {}^3E$ -like internal transitions) of the four VV° centers at $T = 4$ K and their simulation with Gaussian line shapes (left) [(a), (c), (e), (g)]; detail of the ZPL spectra displaying a doublet structure (right) [(b), (d), (f), (h)].

and kh considerably smaller anisotropies $g_{//} - g_{\perp}$ are predicted by DFT, but still with almost identical $g_{xx} \approx g_{yy}$ values (cf. Table I), showing that the deviation from axial symmetry of the spin system is in fact rather small (hk) or even negligible (kh).

The magnetic anisotropy of the spin systems is also reflected by the zero-field splitting (ZFS). For the axial pairs the agreement with experimental values is very well, allowing an attribution of the divacancy configuration to the EPR spectra also for the low-symmetry centers. For the basal pairs,

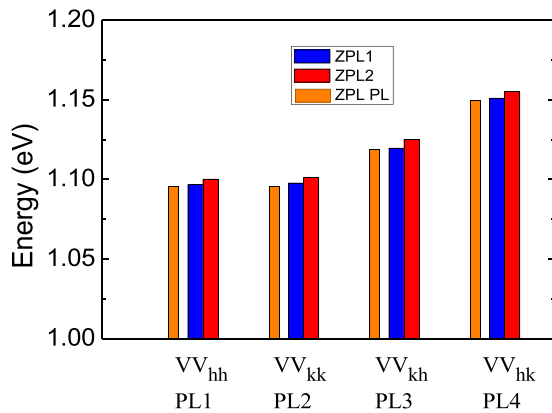


FIG. 9. Experimental values of the doublet-split ZPL lines of the VV centers in 4H-SiC (ZPL1 blue, ZPL2 red; this work) and (orange) ZPL measured by PL [7]. The doublet-splitting is in the 4 meV regime (3.6, 3.4, 5.4, 4.1 meV for PL1, PL2, PL3, PL4, respectively).

the values are underestimated by about 5%, but thanks to the anisotropy parameter E , we can conclusively discriminate between hh and kh. With $E = 89$ MHz the latter provides about twice the anisotropy of the hh configuration (46 MHz), in nice agreement with experiment (82 and 37 MHz). The experimental E value for the hk pair is somewhat larger as previously reported in literature [7].

In Table II, the D and E parameters are decomposed into contributions from spin-spin interaction (D_{ss}) and spin-orbit interaction (D_{so}). The *direct* D_{ss} values considerably overestimate the experimental values. This fact reflects that DFT gives nonvanishing values also for the magnetic $m_s = 0$ sublevel. In a recent study we have shown that this is related to the general problem of spin-contamination of DFT wave functions [21], which can be estimated by analyzing all possible antiferromagnetic spin-configurations representing the $m_s = 0$ sublevel (see also Fig. 4, right). Correcting for this nonvanishing $D_{ss}(m_s = 0)$ values, i.e., defining an effective spin-spin D -value $D_{ss} = D_{ss}(m_s = \pm 1) - D_{ss}(m_s = 0)$, brings the spin-spin parameters D_{ss} much closer to the experimental data [21,38]. Notably, for the axial divacancies, the spin-orbit contributions D_{so} are able to adjust the resulting total DFT-calculated values to the experimental values. For the basal configurations, in particular for the kh configuration, the D_{so} values are considerably smaller, but still able to reduce the remaining difference between DFT-calculated and experimental ZFS parameters.

B. Results and discussion: Optical resonant excitation

We have shown previously for the case of NV centers in 4H-SiC [15] that the measurement of the ZPL and the phonon side bands via resonant excited EPR spectroscopy is extremely powerful as it allows to separate the ZPL spectra of each defect (hh, kk, kh, hk) without interference from the other configurations. In photoluminescence excitation spectroscopy based on the analysis of the phonon side bands the distinction is more complicated due to the superposition of the different PL spectra. We have thus chosen here the same approach “*resonant excitation*” as used previously for the NV centers in SiC. The ZPL line shapes as well as the

phonon coupling of the divacancy centers in 4H-SiC have been investigated theoretically by different authors [28–32]. The structure of the excited states is equally probed in our approach. It should be noted that the ZPL and higher energy phonon sidebands of the VV centers in 4H-SiC have not yet been investigated by optical absorption spectroscopy experimentally.

Our experimental approach is the following: first, the samples are cooled in dark to $T = 4$ K; under this condition, the VV° centers are not observed as due to the irradiation/annealing modified Fermi-level position below 0.5 eV, they are in the 1+ or 2+ charge state [29]. The predicted charge transition levels and the optical excitation scheme are shown in Figs. 5 and 6. Under optical excitation with wavelength $\lambda < 1200$ nm (> 1.033 eV) the VV° level is populated and below this wavelength the excitation induces in addition the $^3A_2 \rightarrow ^3E$ -like internal absorption. As a consequence of the spin dependent intersystem crossing the VV centers become polarized in the $m_s = 0$ state. To measure the absorption spectra (ZPL, PSB) we fix the magnetic field on the known resonance field of the ZPL line of a particular VV° center and monitor its EPR intensity as a function of the monochromatic excitation, decreasing the excitation wavelength step by step by 3 nm. After each excitation the sample is relaxed in darkness for typically 5 minutes until no VV° signal is any longer observed.

The results are shown in Fig. 7. At excitation wavelengths above the ZPL threshold the particular VV° center is at zero intensity. Decreasing the wavelength step by step we measure the EPR signal intensity at each excitation wavelength, which reflects the optical excitation spectrum, characterized by the ZPL absorption line and the associated phonon sidebands at shorter wavelength of the VV centers. The resolution of 3 nm allows revealing a small doublet splitting of the order of 4 nm of the ZPL absorption lines for all four VV centers (Table II). The phonon sidebands of the basal VV_{kh} and VV_{hk} centers are well resolved and have been fitted with Gaussian lines. They show a dominant phonon coupling of 20 and 40 meV.

In Fig. 8, we show the same results plotted on an energy scale, which allows comparison with recent theoretical work [28]. In the energy range explored here, 100 meV from the ZPL energy, we observe a smoothly increasing absorption as predicted in [25]. In the case of the PL3 and PL4 centers, phonon sidebands $E_0 + 24$ meV, $E_0 + 35$ meV, $E_0 + 47.5$ meV (VV_{kh}), and $E_0 + 11.5$ meV, $E_0 + 18.6$ meV (VV_{hk}) are resolved. The ZPL and the phonon coupling of the VV° centers have been studied by various authors. Using density function theory (DFT) in combination with the generating function approach [32], the ZPL and phonon-side bands (PSB) of the VV_{hh} and VV_{hk} centers have been calculated. They identified the PSB centered at 23, 35, and 48 meV to originate predominantly from several delocalized bulk modes. The 23 meV mode has been assigned to a planar optical E_2 mode and the mode at 35 meV to a mixed symmetry of planar acoustic and optical modes with symmetry E_1 . The Debye-Waller factors are in the range of 5% [28].

In Fig. 9, we compare the energies of the absorption ZPL determined in this work by photo-EPR with those previously obtained from photoluminescence spectroscopy [7]. As expected, the values are identical within experimental error, but

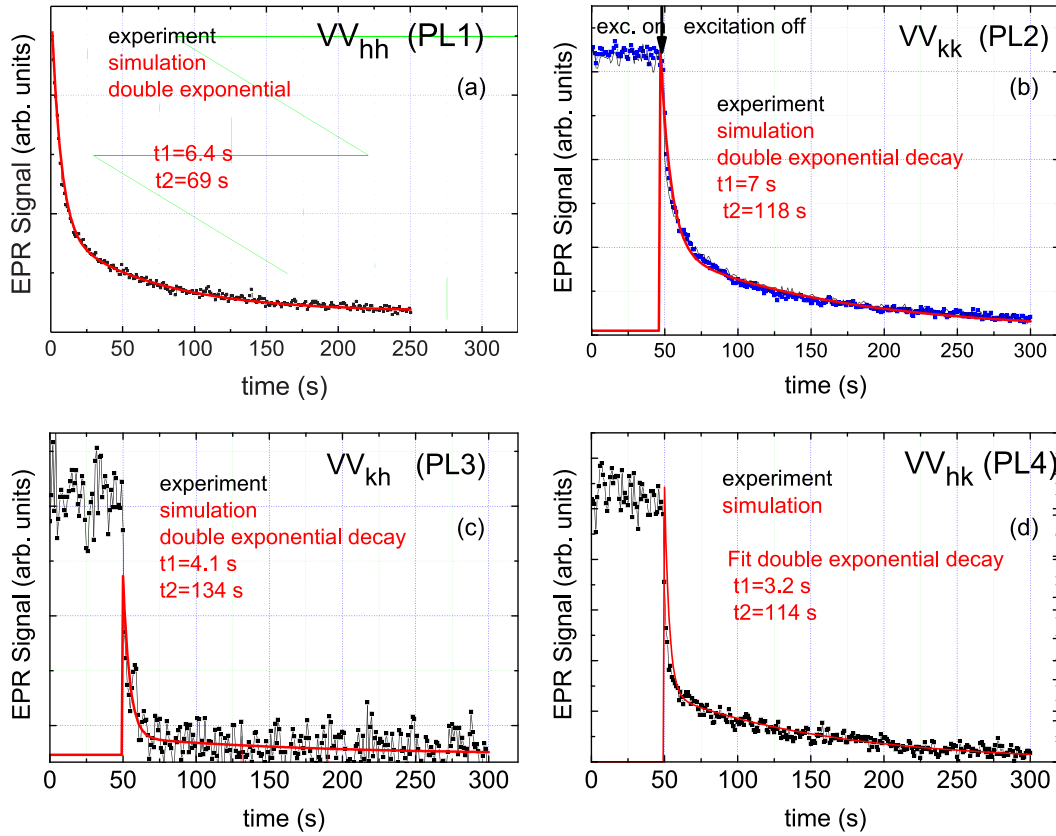


FIG. 10. Decline of the $VV^{\circ} {}^3A_2$ ground-state population (black) after cutoff of the optical excitation and simulation with a double exponential decline (red) for VV_{hh} (a), VV_{kk} (b), VV_{kh} (c), VV_{hk} (d).

the PL measurement did not show a doublet-splitting of the associated ZPL. In a previous resonant EPR work, however, a similar doublet splittings has been already observed for the NcV_{Si} - centers in $4H$ -SiC, but notably only for the axial pairs [15]. The origin of these doublet splittings is still unclear. According to literature, the splitting appears to be clearly too small to be explained by Jahn-Teller distortions [28], but too large to be due to spin-orbit splitting and strain [33,34]. Most probably, several effects have to be taken into account, also the fact that in the present study—in contrast to other

PL investigations—the ZPL are measured in a considerable magnetic field of 0.3 T. A final explanation requires further extensive experimental and theoretical investigations and remains for future work.

C. Low temperature dynamics of the optical induced electron spin polarization

We have equally investigated the charge-state stability and the dynamics of the optical induced spin polarization for the

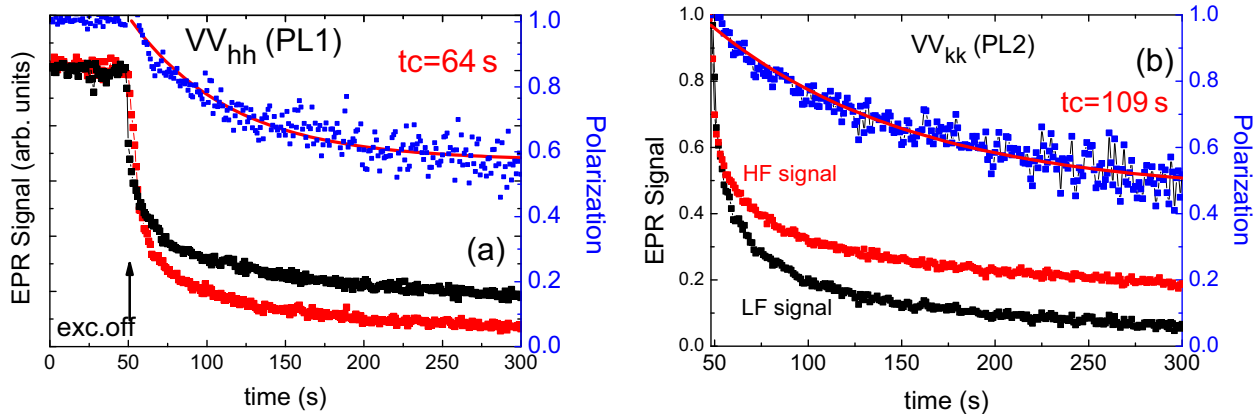


FIG. 11. Decline of the spin polarization of the VV_{hh} (PL1) (a), VV_{kk} (PL2) (b) centers at $T = 4$ K; the low (LF, black) and high (HF, red) field lines remain in absorption/emission demonstrating spin polarization for times ($t > 300$ s) superior to the hole capture; the decline of the polarization can be fitted with a single exponential with a time constants $t_c = 64$ s and $t_c = 109$ s, respectively.

four VV° centers. The VV centers were excited with a 808 nm laser diode with a power of 100 mW. The time resolution of our setup is 100 ms. The intensity of the VV° EPR spectra (monitored for the low field lines of the $S = 1$ centers) saturates rapidly with a time constant below 100 ms and remains stable for at least 10 min. This demonstrates a high charge-state stability under optical excitation, which is not observed in initially n -type conducting N doped samples. Here, due to the optically induced population of the VV^- charge state with a $(-/0)$ charge transition level at about 1.2 eV above the $(0/+)$ level continuous repumping is necessary to repopulate the neutral VV° charge state [35–37]. The quenching is mainly related to the photoionization of VV° by additional defects such as V_C , V_{Si} and N donors, which are still present in N doped samples after irradiation and annealing. In p -type material the charge transition level $(0/+)$ is the relevant one; here the transition $VBM \rightarrow VV^\circ$ have slightly larger energy thresholds than the intracenter excitations, which make the present experiments possible in p -type material.

As shown in Fig. 10 and Table III, the decline of the neutral VV° states at $T = 4$ K due to hole capture is in fact very slow with time constants up to hundreds of seconds. The decline can be fitted with a double exponential with two characteristic decay times of $t_1 = 5$ s and $t_2 = 100$ s, slightly varying with the VV center configuration.

The observed charge-state stability has strong implication for the application of the VV centers as qubits, as the dynamics of the optically induced electron spin polarization is a prime parameter. In n -type $4H$ -SiC, the excitation of the $^3A_2 \rightarrow ^3E$ -like internal transition ionizes simultaneously the VV° centers by emitting an electron into the conduction band, so that repumping procedures are required to initialize the spin polarization. The present case of p -type material is much more favorable as the optical excitation leads rapidly to a metastable 100% spin polarization which is limited only by the spin-lattice relaxation time T_1 as shown in Fig. 11.

The optically induced spin polarization has been analyzed theoretically previously [38,39]. The buildup of the spin polarization in our p -type sample is fast as compared to the time resolution of 100 ms in the EPR experiment and cannot be analyzed. We observe a $P = 100\%$ polarization after 100 ms photo excitation. The decline of the spin polarization after cutoff of the optical excitation could be easily measured as it persists at a timescale of minutes at $T = 4$ K. The result is shown in Fig. 11 for the VV_{hh} , VV_{kk} centers. It is obtained by the measurement of the decline of the intensities of the respective low-field and high-field lines of each VV center.

It should be noted that the polarization declines much slower than the hole capture and the phases of the two lines persist in absorption and emission, respectively. For an easier visualization the intensities of the low-field and high-field EPR lines I_1 , I_2 are superposed, i.e., the high-field line values have been multiplied by a factor of -1 . The polarization is calculated as $P = (I_1 - I_2)/(|I_1| + |I_2|)$. The decline of the polarization can be fitted with a single exponential decay. Very similar values have been obtained for the other two VV centers.

IV. CONCLUSIONS

We have investigated the spin and optical properties of the VV° centers in Al doped p -type $4H$ -SiC by low temperature X-band Photo-EPR. The ZPL and their phonon sidebands (PSB) have been determined separately for each VV center via their EPR signal intensity. Our results support the previous assignment of the PL1, PL2, PL3, PL4 photoluminescence spectra to the hh, kk, kh, hk configurations of the VV centers. Surprisingly, we observe a second ZPL for each VV center, also for the basal configurations, resulting in a 4 meV-split doublet structure, which had not been reported before. For the basal VV centers we resolved phonon sidebands at $E_0 + 24$ meV, $+36$ meV, and $+47.5$ meV (V_{kh}) and $E_0 + 11.5$ meV, $E_0 + 18.6$ meV (VV_{hk}). The optical excitation in p -type material polarizes the VV centers rapidly to $P = 100\%$ in the $m_s = 0$ state in spite of the low intensity of the light source. The electron spin polarization saturates within $t \ll 100$ ms and declines after switchoff of the excitation in a single exponential decay with a time constant of about 100 s at $T = 4$ K. Contrary to the case of VV centers in nitrogen doped n -type $4H$ -SiC, the spin polarization is metastable under continuous optical excitation, not requiring an optical repumping procedure.

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DATA AVAILABILITY

The data that support the findings in this Article are available from the authors upon reasonable request.

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